

Question

The reddish-brown element **X**'s allotrope **Y** is oxidized by O_3 at 195 K to produce an orange oxide **A**, and further oxidation yields a colorless oxide **B**. It is known that the mass fractions of oxygen in **A** and **B** are 0.2310 and 0.3336 respectively, and the oxidation states of element **X** in **A** are different, and the **X** atoms are not adjacent.

Y can react with SbF_5 in XF_5 solvent to obtain an ionic compound **C** with a single charge on both the cation and anion. The coordination number of all Sb atoms in **C** is 6. The cation has a linear structure, the mass fraction of Sb in the anion is 0.5458, the anion does not contain element **X**, and the cation does not contain element F.

Regarding the above compounds, the correct statement is:

1. The oxidation states of element **X** in **A** differ by 4.
2. The chemical environments of element **X** in **B** are different.
3. The number of element **X** in the chemical formulas of **A** and **B** are different.
4. The cation in **C** is composed of only one element.
5. The element composition ratio in the chemical formula of the anion in **C** is 4 : 21.

A. 1, 2

B. 1, 3

C. 1, 4

D. 1, 5

E. 2, 4

F. 2, 5

G. 3, 4

H. 4, 5

Answer

Correct Answer: **C**

Detailed Explanation

Based on the reddish-brown element **Y** and the orange oxide **A** and colorless oxide **B**, it can be inferred that **X** is the element Br; **Y** is the element Br₂; **A** is Br₂O₃; **B** is Br₂O₅. The calculated mass fraction of oxygen element is consistent with the question, which can be verified.

CHECKPOINT

1 PTS

X is the element Br

CHECKPOINT

1 PTS

Y is the element Br₂

CHECKPOINT

1 PTS

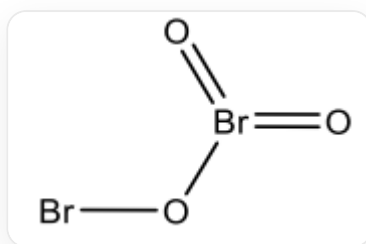
A is Br₂O₃

CHECKPOINT

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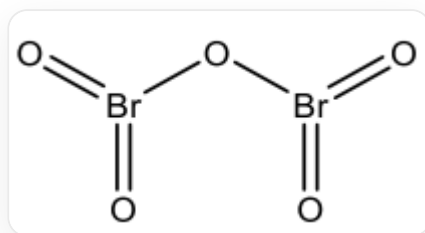
B is Br_2O_5

The structure of **A** is:



Structure of Br_2O_3 : One Br is connected to O through two bromine-oxygen double bonds and one bromine-oxygen single bond; the other Br is connected to O through a bromine-oxygen single bond; two Br are separated by one O

The structure of **B** is:



Structure of Br_2O_5 : Two Br are each connected to O through two bromine-oxygen double bonds and one bromine-oxygen single bond; two Br are separated by one O

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In **A**, one Br is connected to O through two bromine-oxygen double bonds and one bromine-oxygen single bond; the other Br is connected to O through a bromine-oxygen single bond; two Br are separated by one O

CHECKPOINT

1 PTS

In **B**, two Br are each connected to O through two bromine-oxygen double bonds and one bromine-oxygen single bond; two Br are separated by one O

Therefore, statement 1 is correct; statements 2 and 3 are incorrect.

It is known that the anion in **C** does not contain the Br element and contains the Sb element, so it can be known that the anion is composed of the Sb element and the F element. A typical anion family is **Double subscripts: use braces to clarify**. When $m = 3$, it meets the mass fraction of the Sb element in the question.

Therefore, the anion is $[\text{Sb}_3\text{F}_{16}]^-$.

According to the cation not containing the F element and having a linear structure, it can be deduced that the cation is $[\text{Br}_2]^+$.

C is $[\text{Br}_2]^+[\text{Sb}_3\text{F}_{16}]^-$.

CHECKPOINT

1 PTS

C is $[\text{Br}_2]^+[\text{Sb}_3\text{F}_{16}]^-$

Therefore, statement 4 is correct and statement 5 is incorrect.

Select option C.